

Optimizing UMD Transit Operations

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Project Proposal:

This project proposes a Data Warehousing solution to address the operational inefficiencies of the University of Maryland's Shuttle-UM system (DOTS), specifically regarding the impact of external factors like football games at SECU Stadium, basketball games at Xfinity Center, and inclement weather. By integrating simulated historical trip logs for key routes (like the 104 College Park Metro and 105 Campus Connector) with campus event calendars and local weather data, we aim to analyze how these high-traffic events correlate with service delays and overcrowding. The final deliverable will enable DOTS to move from reactive to predictive scheduling, answering critical business questions about when to deploy relief buses during surges and how specific weather patterns disrupt standard route completion times.

1. Business Problem Description The University of Maryland's Department of Transportation Services (DOTS) operates a critical transit network serving over 3 million passengers annually. While standard schedules function adequately during average weeks, the system frequently experiences severe service degradation including overcrowding, "bunching" (multiple buses arriving at once), and significant delays during high-stress scenarios. These scenarios are primarily driven by external factors such as major athletic events (e.g., football games at SECU Stadium), inclement weather, and peak academic activity like the midterms, career fairs, and finals.

Our team proposes that DOT would benefit from an integrated analytical view to evaluate these external variables can impact route performance. The business problem we are addressing is the inability to predict service disruptions caused by these non-standard days. By moving from a reactive to a data-driven predictive model, DOTS could optimize resource allocation by moving service to more high-demand areas and navigate shifting needs.

2. Data Sets for Integration We will integrate three distinct data sources to construct our Data Warehouse:

- **Source A: Shuttle-UM Operations Data (Simulated/API Extract)**

- Source: Extracted via the **umd.io API** (Routes/Stops) and **<https://transitapp.com/partners/apis>** and simulated for historical transactional volume
- Description: Granular trip logs representing bus locations and passenger loads.
- Key Fields: Trip_ID, Route_ID (104, 105), Bus_ID, Stop_ID, Scheduled_Arrival_Time, Actual_Arrival_Time, Passenger_Count.
- **Source B: Campus Event & Athletics Calendar**
 - Source: **umterps.com** and **calendar.umd.edu**.
 - Description: Structured schedule of high-impact campus events.
 - Key Fields: Event_ID, Date, Event_Type (Football, Basketball, Commencement), Venue (SECU Stadium, Xfinity Center), Expected_Attendance.
- **Source C: Local Weather Archive**
 - Source: **NOAA / Weather Underground (Station: KCGS)**.
 - Description: Historical hourly weather conditions for College Park (Zip 20742).
 - Key Fields: Date_Hour, Condition (Rain, Snow, Clear), Temperature_F, Wind_Speed, Precipitation_Inches.

3. Target Data Model (Star Schema) We propose a **Dimensional Model** centered around a **Fact_Shuttle_Trip** table to facilitate high-performance querying:

- **Fact Table:** Fact_Shuttle_Trip (Grain: One row per bus arrival at a specific stop).
- **Dimensions:**
 - Dim_Route (Route Name, Loop Type).
 - Dim_Stop (Stop Name, GPS Coordinates, Shelter Available Y/N).
 - Dim_Date & Dim_Time (Standard temporal dimensions).
 - Dim_Weather (Hourly weather conditions linked to trip time).
 - Dim_Event (Campus events active during the trip window).

4. Data Quality & Transformation Plan

- **Quality Assessment:** We anticipate challenges with "Missing Data" in the GPS logs (dropped signals) and "Granularity Mismatches" between hourly weather data and minute-by-minute bus arrivals.
- **Transformation (ETL):**
 - **Standardization:** Convert all timestamps to a standard UTC-5 (EST) format to align disparate sources.

- **Cleansing:** Impute missing Passenger_Count values using historical averages for that time slot.
- **Integration:** Map specific Date and Time keys from the Event and Weather datasets to the central Fact table during the loading process.

5. Proposed Analytical Questions The final Data Warehouse will support queries to answer:

1. **Weather Impact:** What is the correlation between precipitation intensity (inches/hour) and the average delay on the 104 College Park Metro route?
2. **Event Latency:** How does the average round-trip time for the 105 Campus Connector change on days with home basketball games (Xfinity Center) vs. non-game days?
3. **Capacity Planning:** Which 30-minute time slots (e.g., 9:00 AM - 9:30 AM) consistently show "Maximum Capacity" status across all routes?
4. **Route Efficiency:** Which specific bus stops have the highest variance between "Scheduled Arrival" and "Actual Arrival" times?